Core ML & Computer Vision Research

Project Description

This project delivers a **research-driven Computer Vision and Deep Learning platform** focused on:

- Model architecture experimentation
- Image processing standardization
- Custom model development
- Deployment pipelines using **PyTorch** and **TensorFlow Serving**

The platform supports the complete lifecycle:

- Data Collection
- Image Augmentation
- Model Architecture Research
- Training
- Validation
- Testing
- Performance Benchmarking
- Deployment
- Monitoring

The primary objective is to build **high-performance, production-ready computer vision models** while conducting structured research on **custom CNN architectures and transformer-based vision models**.

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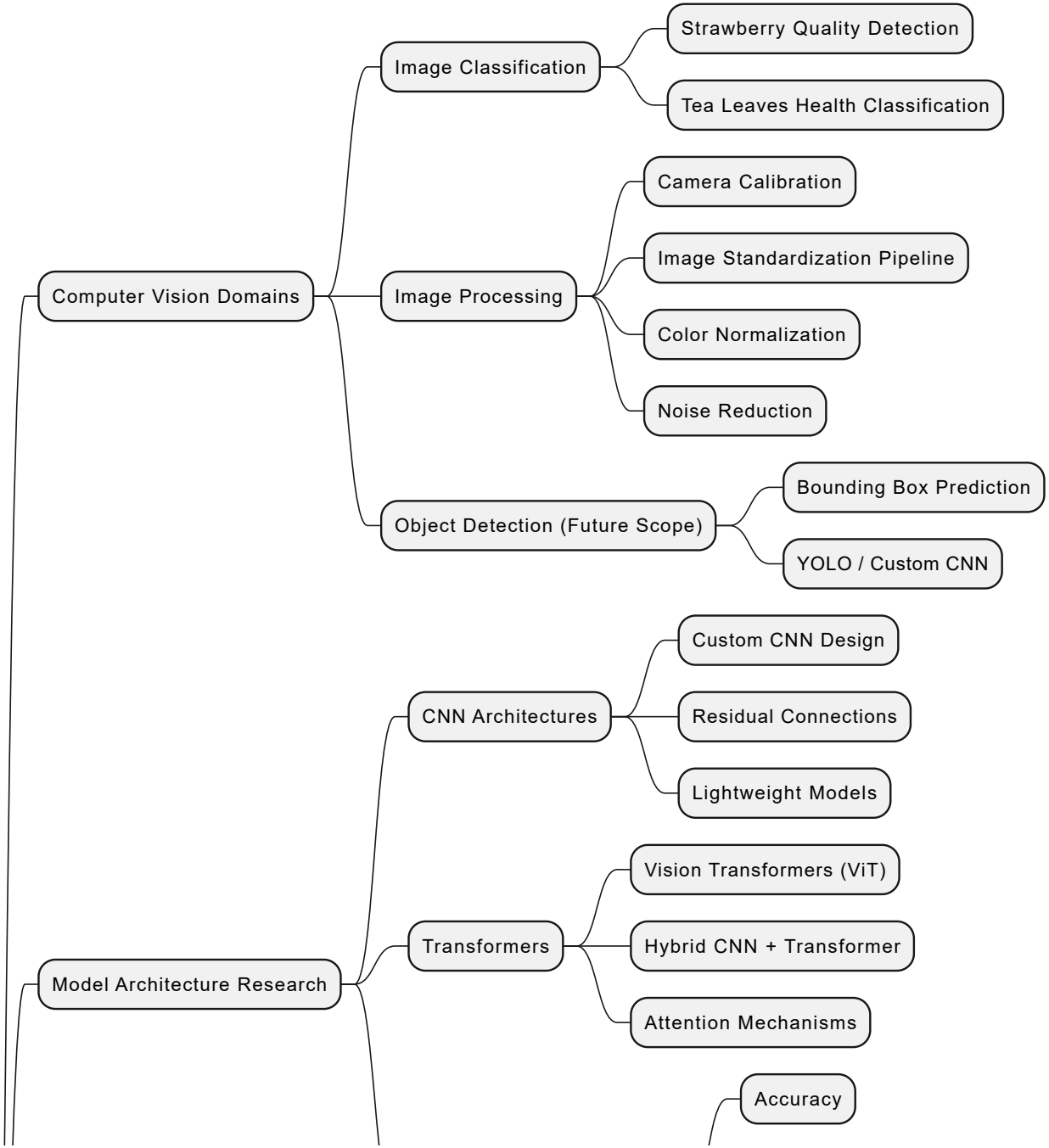
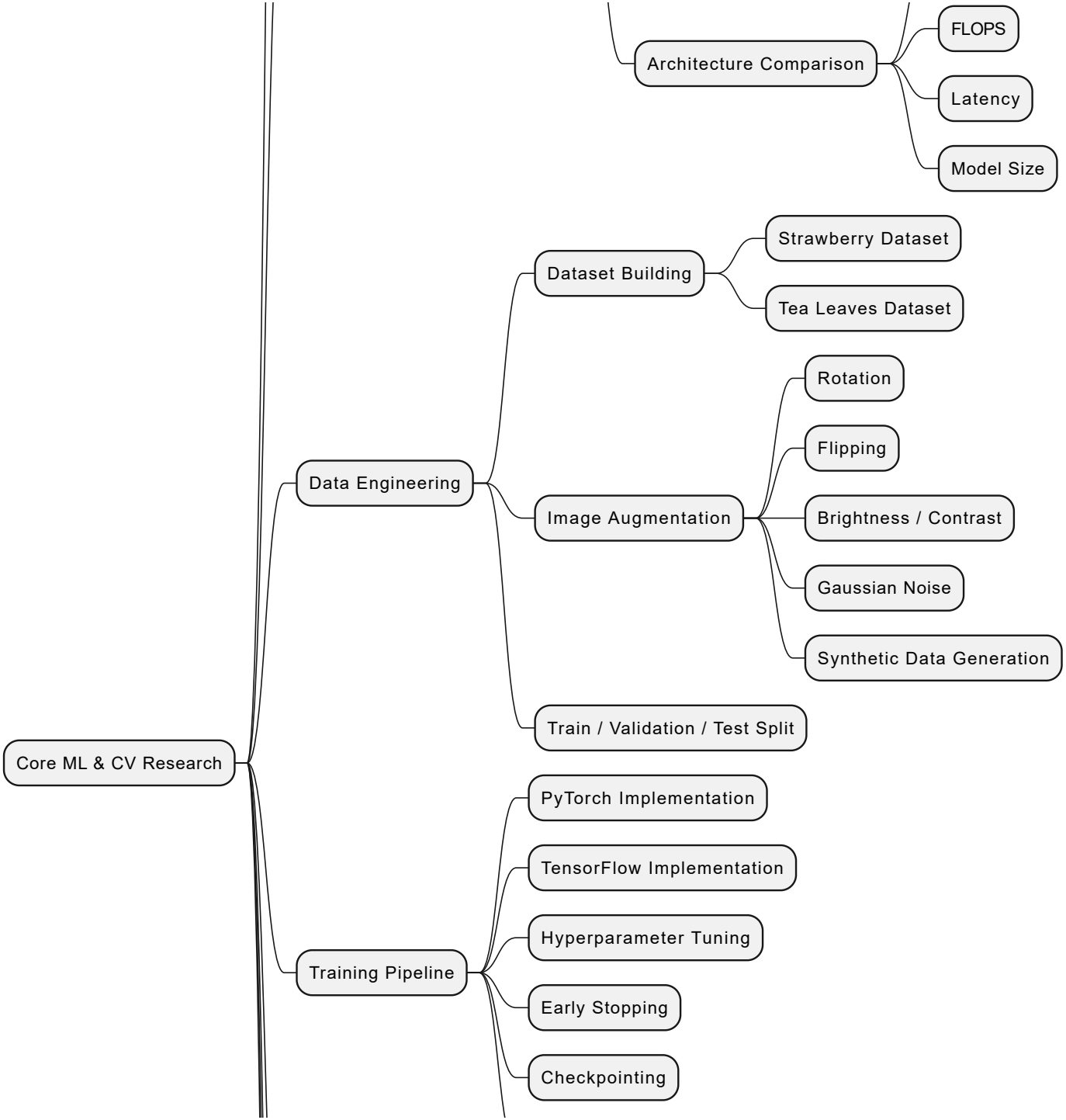
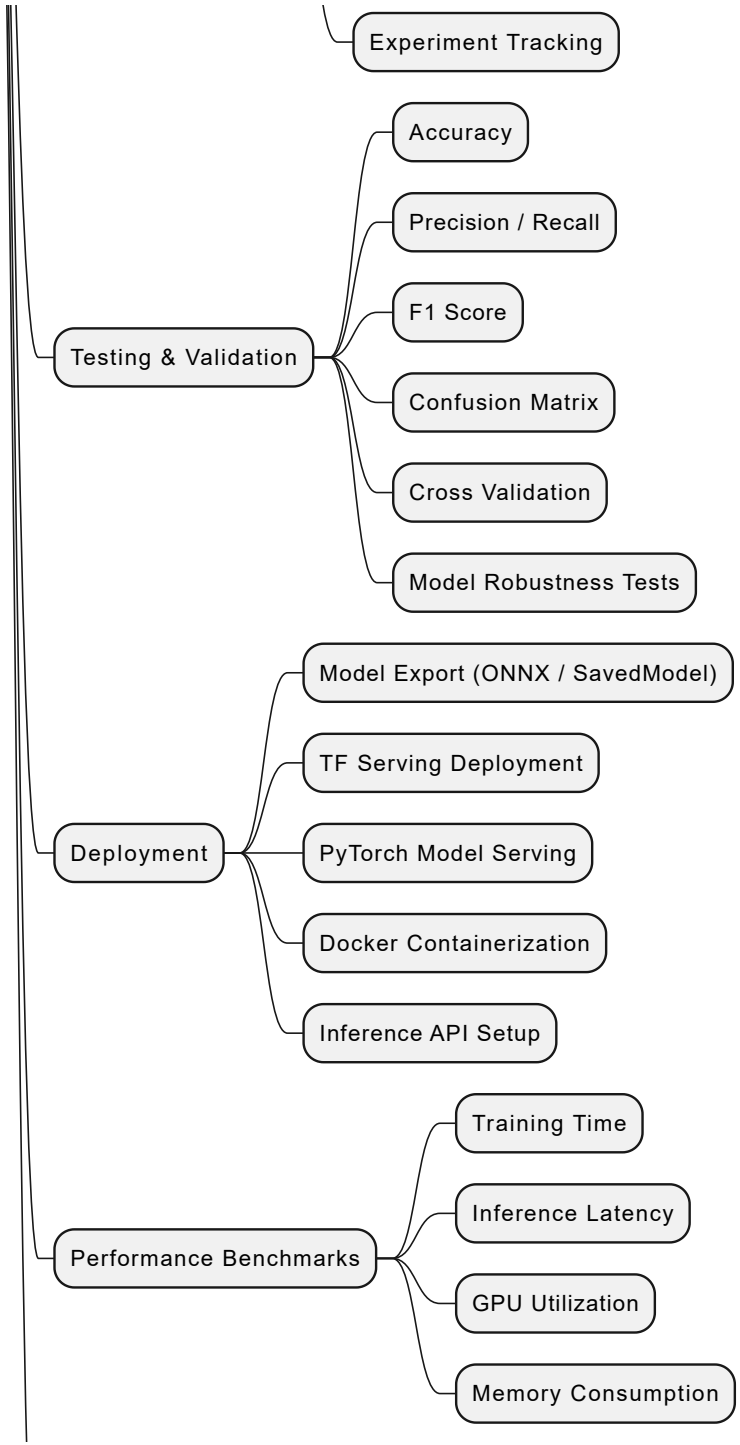
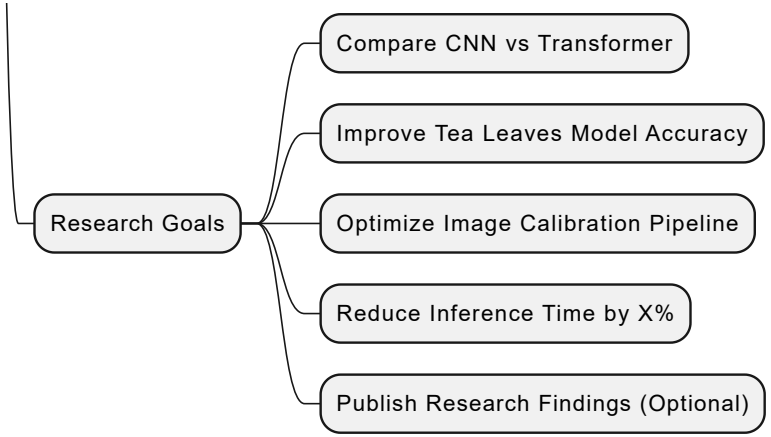



Architecture Comparison
FLOPS
Latency
Model Size
Dataset Building
Strawberry Dataset
Tea Leaves Dataset
Data Engineering
Rotation
Flipping
Image Augmentation
Brightness / Contrast
Gaussian Noise
Synthetic Data Generation
Train / Validation / Test Split
Core ML & CV Research
PyTorch Implementation
TensorFlow Implementation
Hyperparameter Tuning
Training Pipeline
Early Stopping
Checkpointing


Experiment Tracking
Testing & Validation
Accuracy
Precision / Recall
F1 Score
Confusion Matrix
Cross Validation
Model Robustness Tests
Deployment
Model Export (ONNX / SavedModel)
TF Serving Deployment
PyTorch Model Serving
Docker Containerization
Inference API Setup
Performance Benchmarks
Training Time
Inference Latency
GPU Utilization
Memory Consumption



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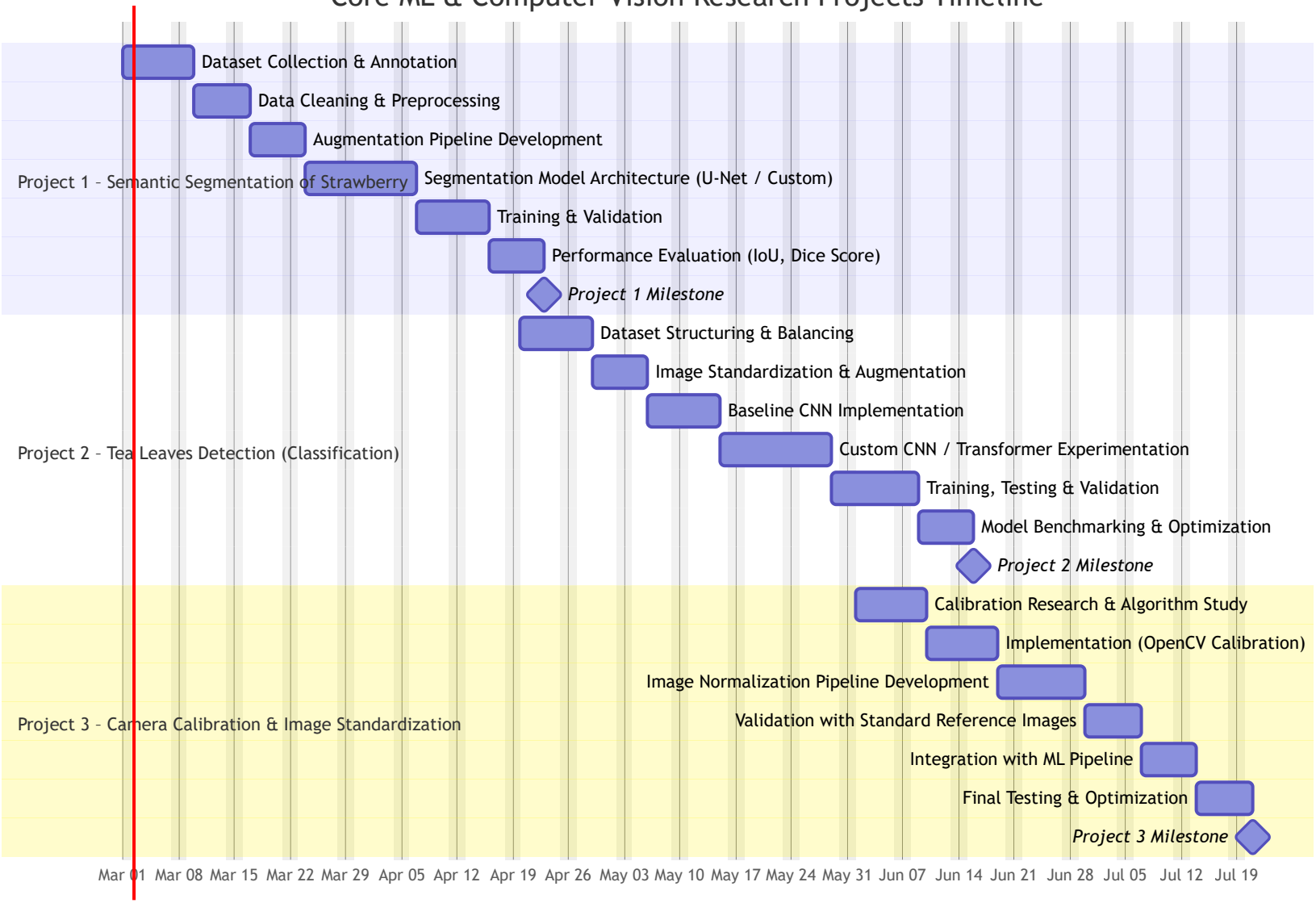
Projects Summary

Attribute	Detail
Project Duration	20 weeks (2026-03-01 to 2026-07-20)
Objective	Build a research-driven CV pipeline with custom architectures, transformer exploration, image calibration processing, and production-grade deployment.
Key Deliverables	Dataset pipelines · Custom CNN architectures · Transformer experimentation · Camera calibration system · Training & testing pipeline · Deployment via TF Serving · Final research report & demo

Core ML & Computer Vision Research Projects Timeline